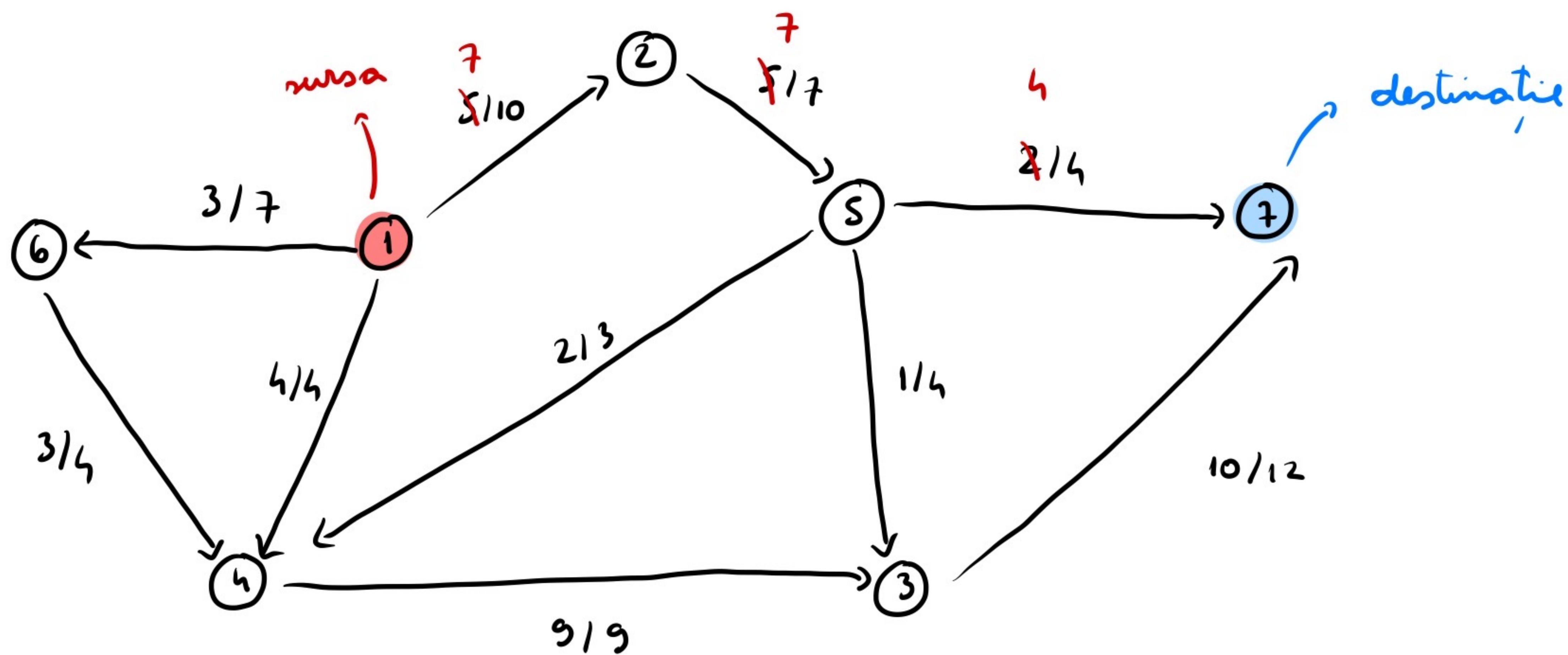


AF Seminar 6

1. Flux maxim
2. Tăietura minimă
3. Havel - Hakimi cu flux
4. Sistem unic de reprezentanți
5. Vertex Cover

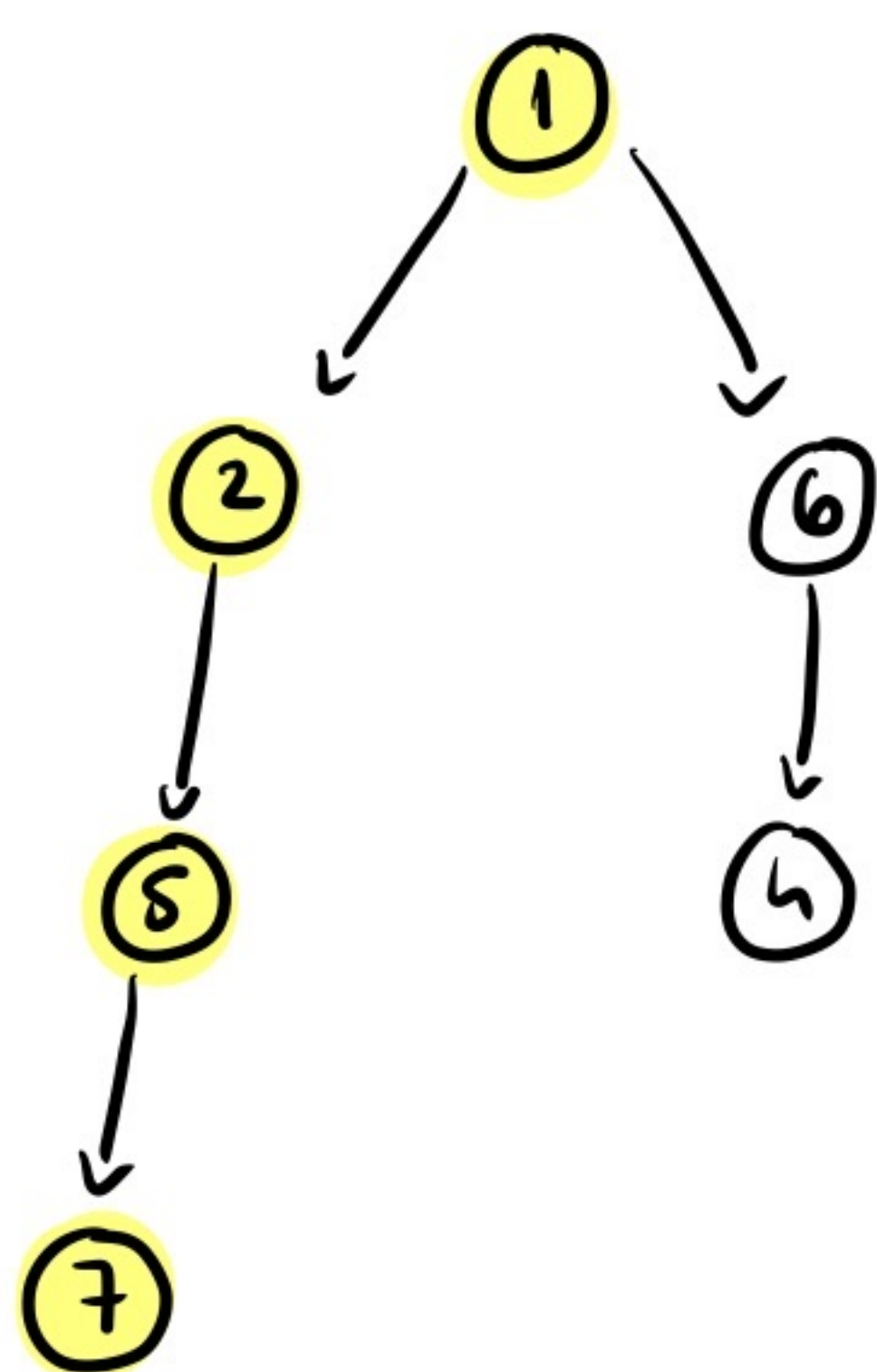
Flux maxim

ex 1



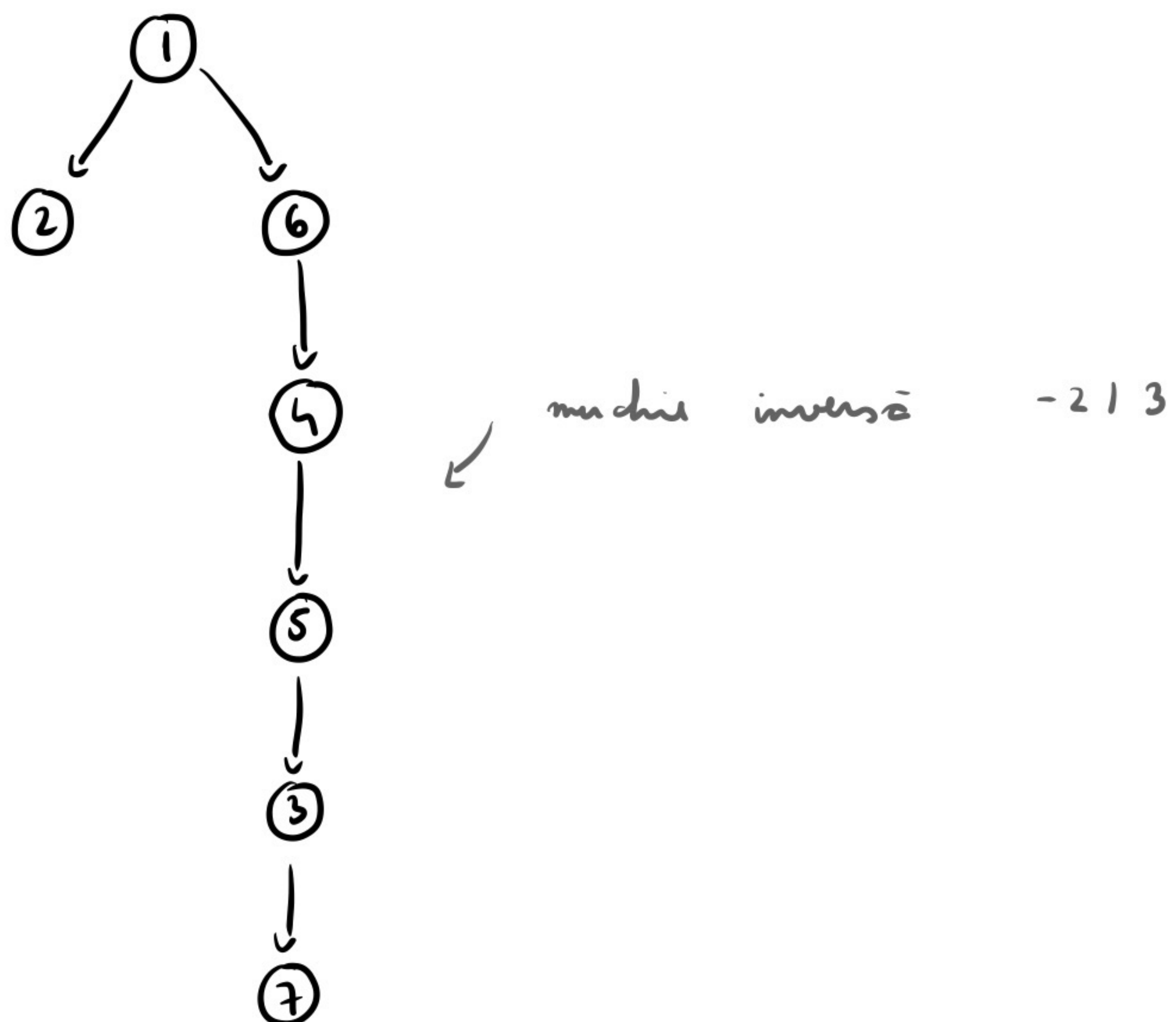
BFS din 1

diferența dintre capacitate și flux > 0



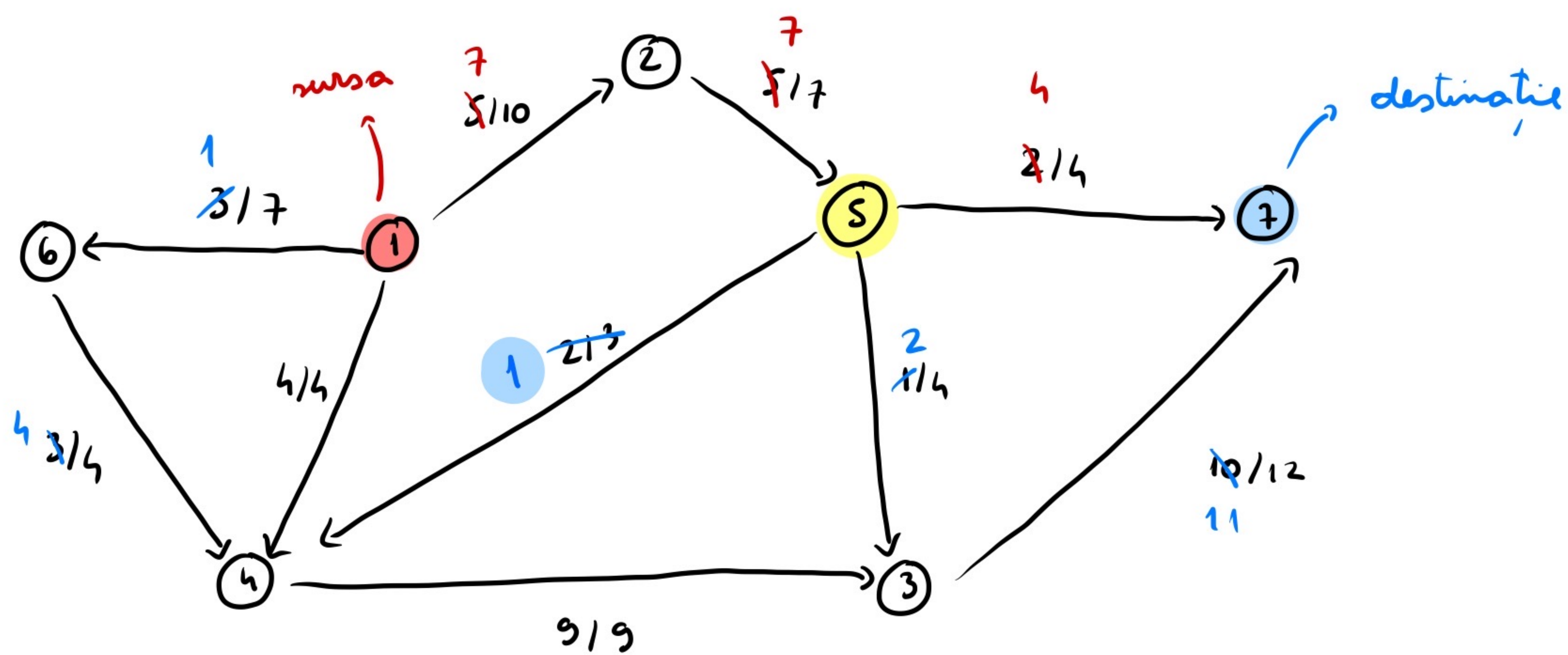
Ce flux putem trimite pe acest drum

1 2 5 7 +2



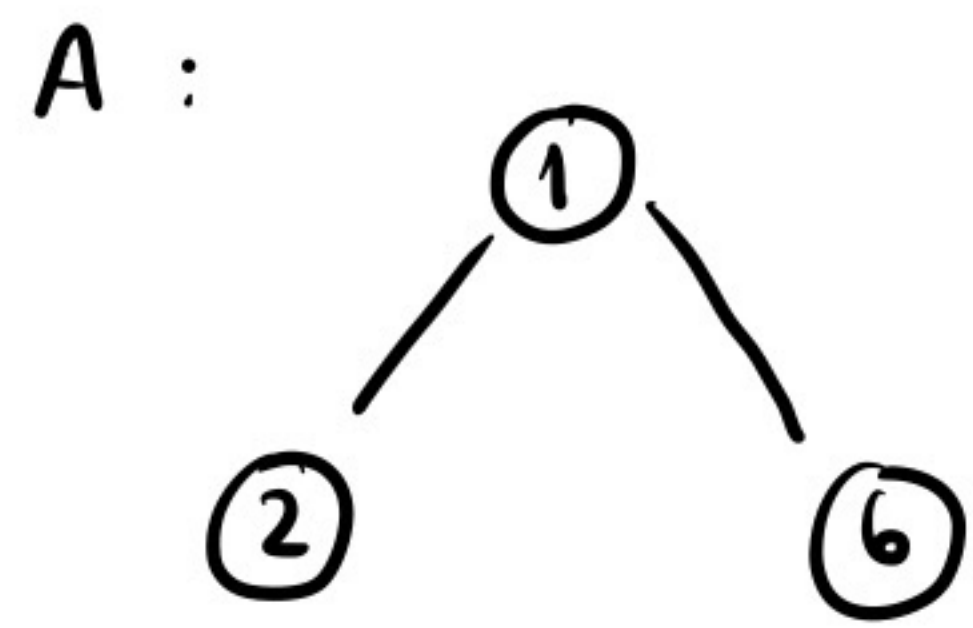
1 6 4 5 3 7 + 1

+1 +1 -1



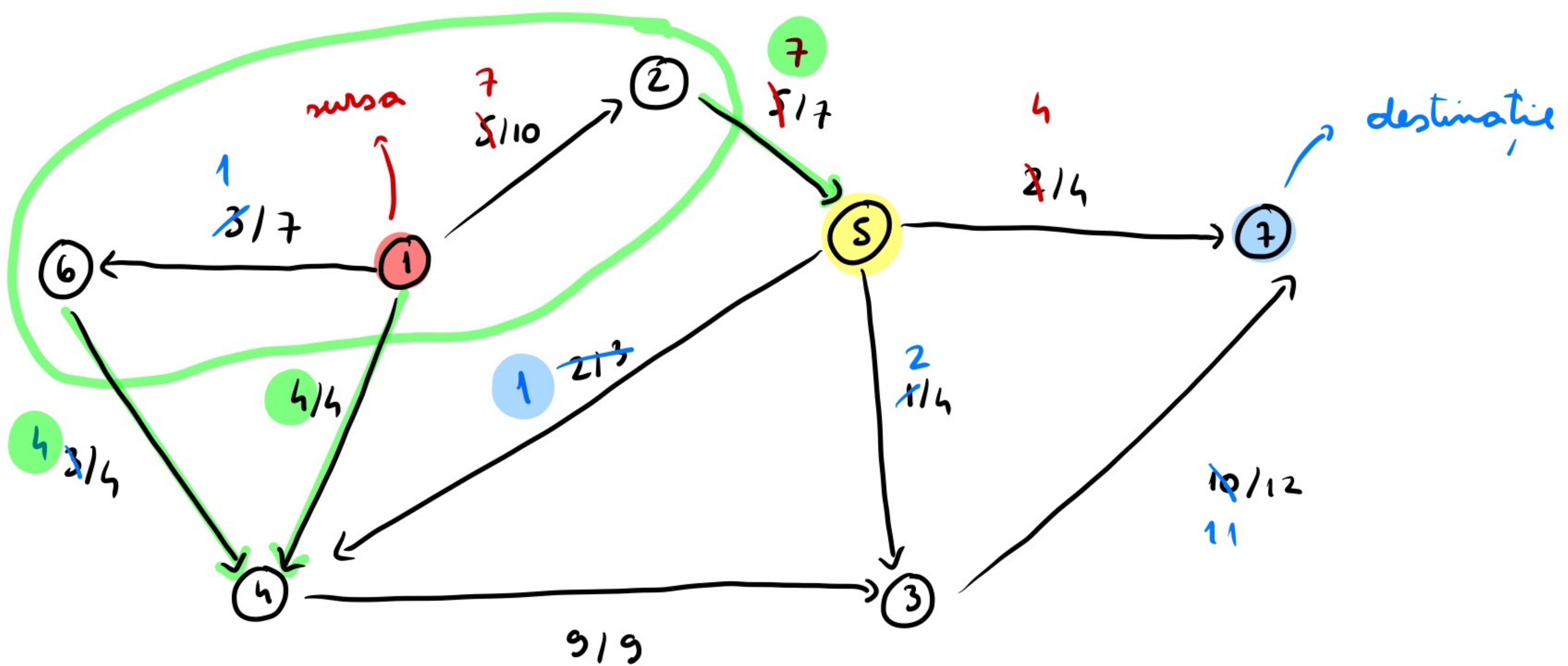
Flux maxim $\rightarrow 15$ (cât flux iese din nursă)

Tăietura minimă



B = toate celelalte

valoarea tăieturii = val flux maxim



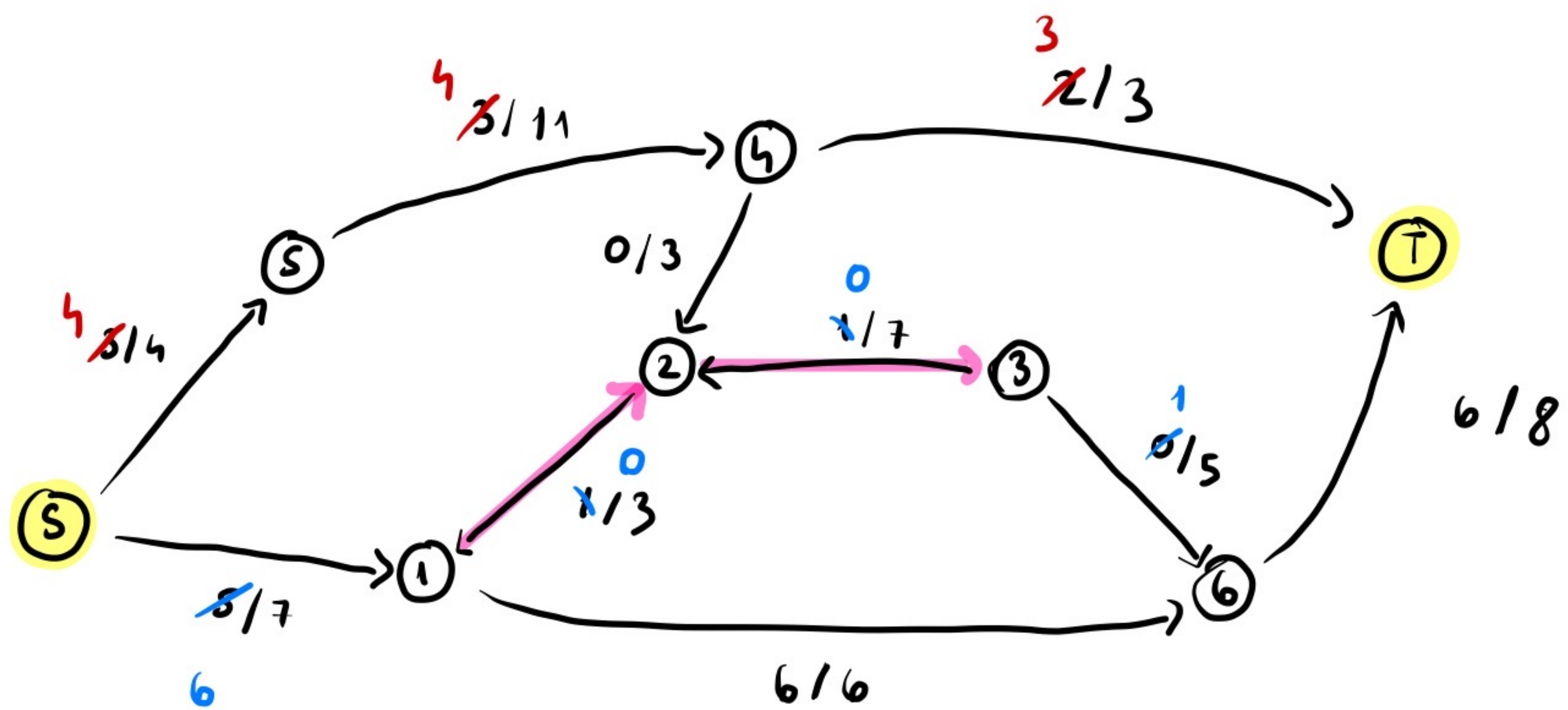
Acele tăieturi = $[6, 4]$, $[1, 4]$, $[2, 5]$

cele inverse se menționează dar nu se adaugă la sumă

A handwritten flow network diagram with 7 nodes (S, 1, 2, 3, 4, 6, T) and 8 edges. The source node S is highlighted in yellow. The sink node T is also highlighted in yellow. The edges and their capacities are: S to 1 (5/7), S to 4 (3/11), 1 to 2 (1/3), 2 to 3 (1/7), 3 to 6 (0/5), 4 to T (2/3), 6 to T (6/8), and 3 to 2 (0/3).

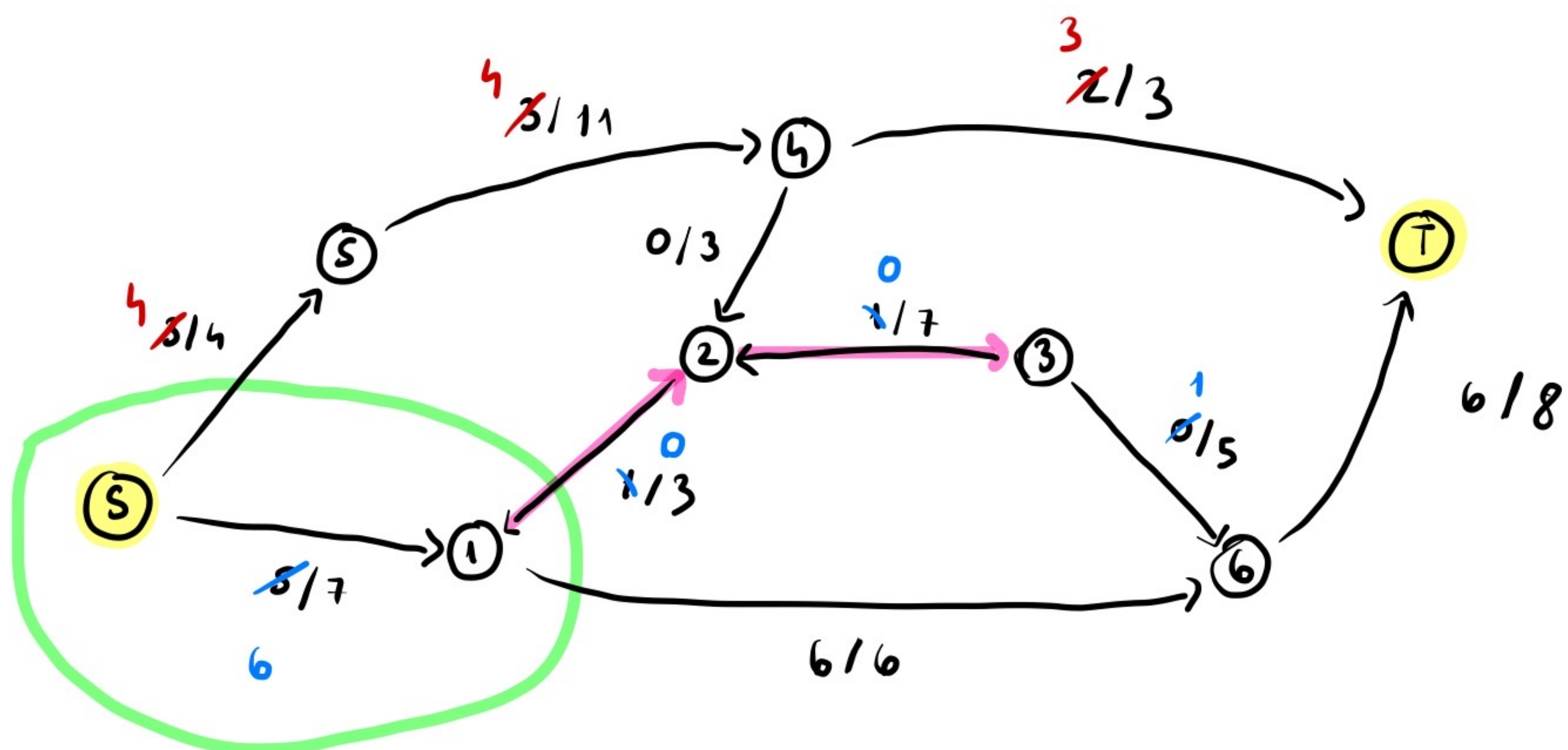
A handwritten graph diagram showing a network flow problem. The graph has 7 nodes: S (source, yellow), T (sink, yellow), 1, 2, 3, 4, and 6. Edges and their capacities are: S to 1 (5/7), S to 4 (4/11), S to T (3/3), 1 to 2 (1/3), 2 to 3 (1/7), 3 to 6 (0/5), 4 to T (6/8), and 6 to T (6/6). Red markings indicate flow adjustments: a red '4' and '3' are crossed out on the S to 4 edge, and a red '3' and '2' are crossed out on the S to T edge.

$S \quad 1 \quad 2 \quad 3 \quad 6 \quad T \quad \rightarrow \quad +1$



Încălzi

BFS din S în acele neșaturate



$A : \{S, 1\}$

$B : \{S, 2, 4, 3, 6, T\}$

Cost încălzire = 10

A ne directe $(S, 5), (1, 6)$

A ne inverse $(2, 1)$

↳ *urechișuri gresite*



Havel - Hahimi in flux

ex 3

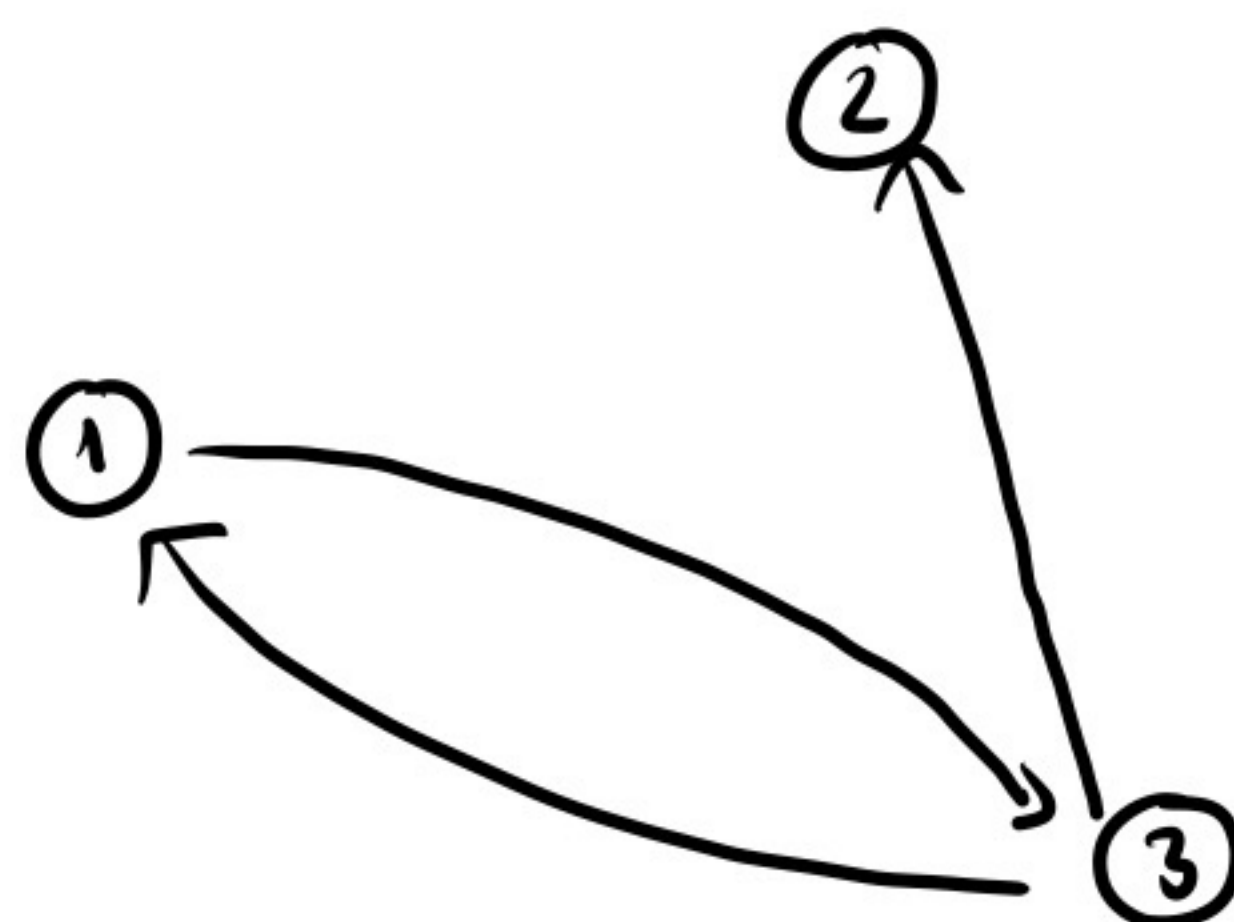
Fie gradele unui graf orientat

+

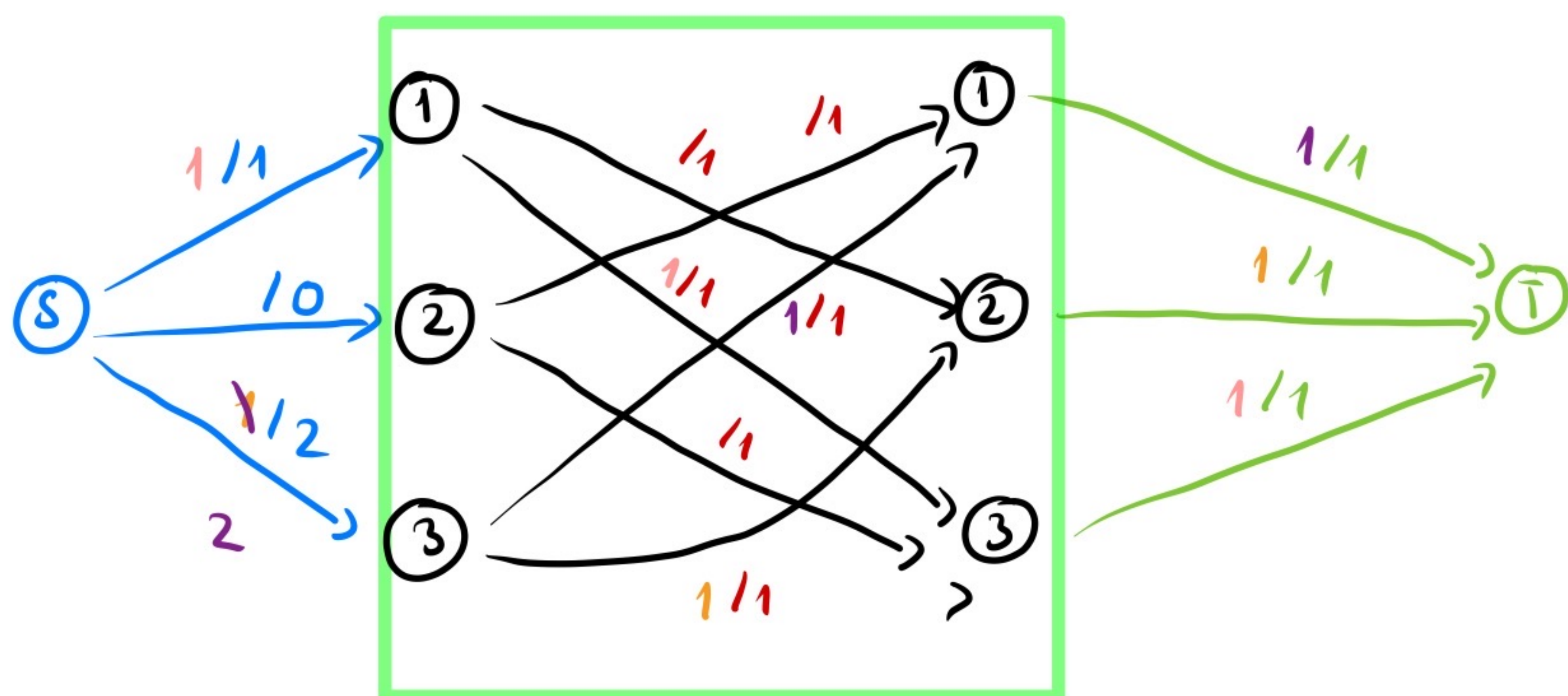
constituiți graful (nu permitem bucle)

grade ieșire
 $d^- = \{1, 0, 2\}$

grade de intrare
 $d^+ = \{1, 1, 1\}$



d^-



d^+

S 1 3 T → +1

S 3 2 T → +1

S 3 1 T → +1

Flux max = suma gradelor astfel nu se poate

Sistem unic de reprezentanți

ex 4 $\{ 1, 2, 3, 4 \}$ valori posibile

$$x_1 = \{ 2, 3 \}$$

$$x_2 = \{ 1, 3, 4 \}$$

$$x_3 = \{ 2, 4 \}$$

+

$$(x_1, x_2, x_3)$$

#

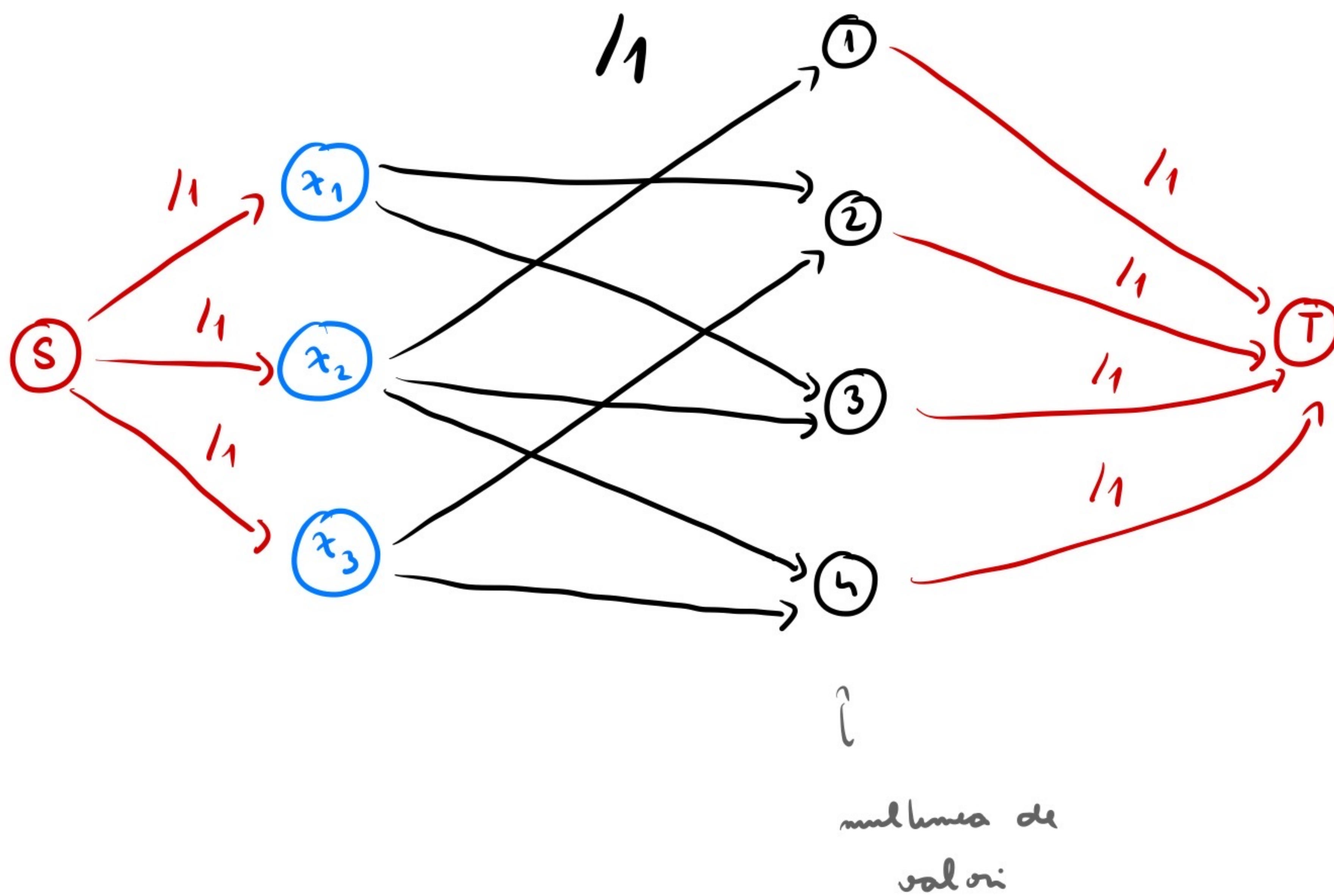
$$x_1 \neq x_2 \neq x_3$$

$$x_1 \in x_1$$

$$x_2 \in x_2$$

$$x_3 \in x_3$$

$$x: (3, 1, 4), (2, 3, 4)$$



nu merge dacă fluxul > nr de mulțimi

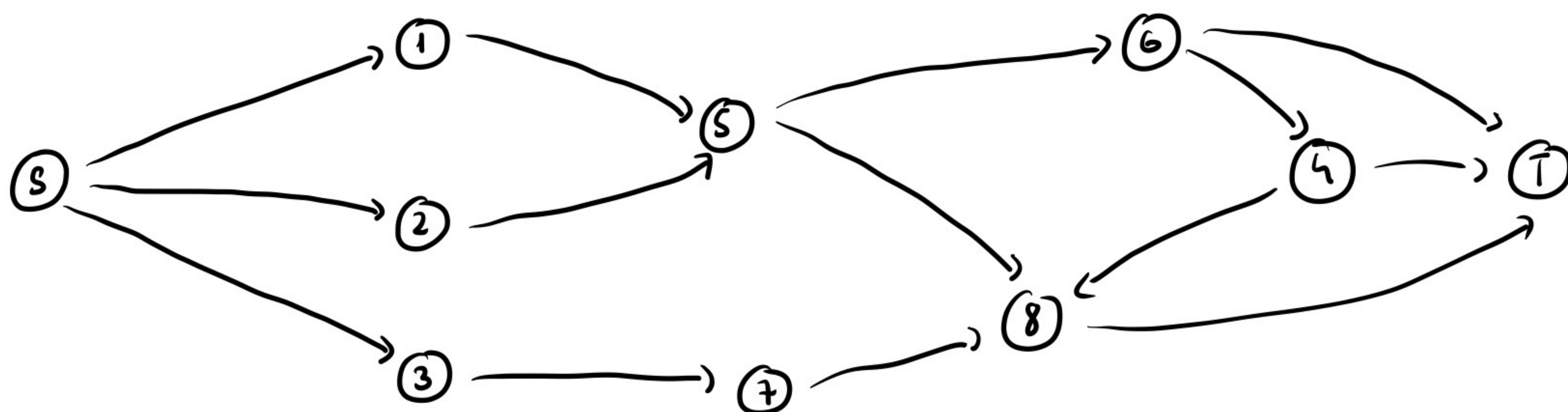
ex 5

6 orientat

+

nr. maxim de drumuri - arc disjuncte de la s la t
#

nu au niciun arc
comun



S 3 7 8 T

S 2 5 6 4 T

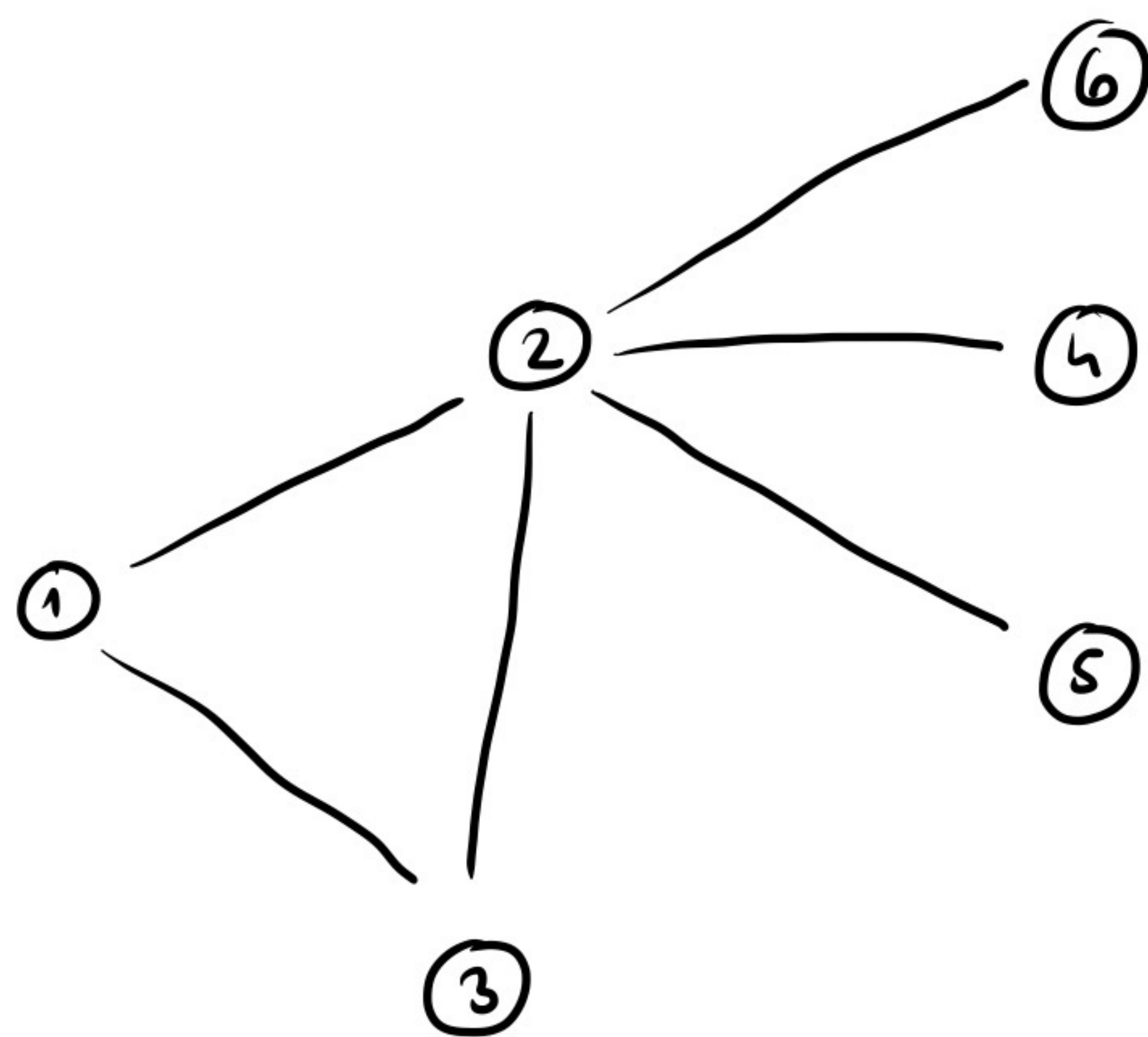
1. Punem pe fiecare arc capacitatea 1

2. Aplicam flux maxim

3. Nr de drumuri = fluxul



Vertex cover



Selecții @ multime de noduri

a.i. toate muchiile să aibă un capăt în acea multime

R: { 2, 1 }

{ 2, 3 }

Grafuri generale $\in$ NP hard

bipartit $\in$ P (polinomial)

T. König

G bipartit

MVC $\Leftrightarrow$ MM

MVC = minimum vertex cover

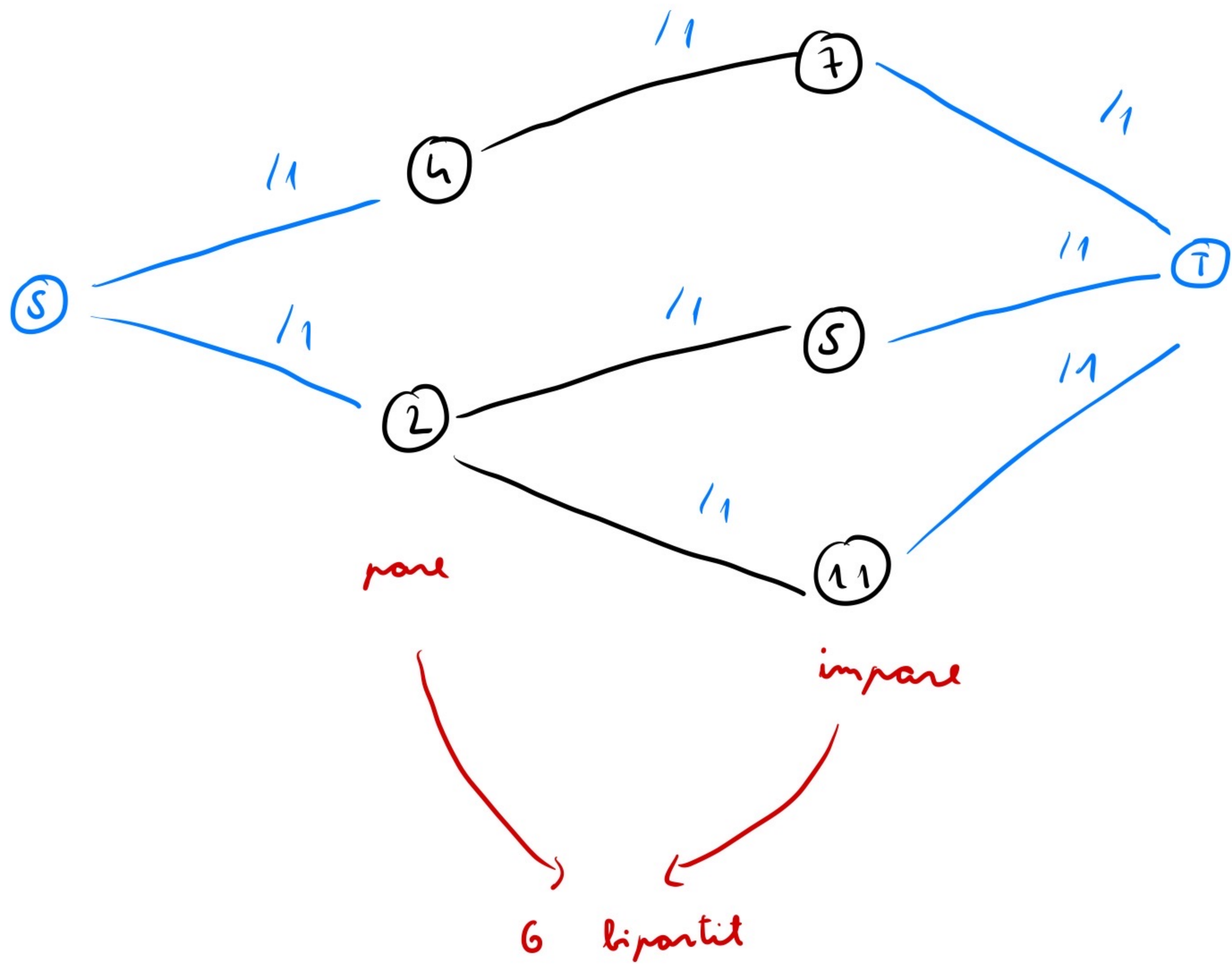
MM = maximum matching (cuplaj)

ex 6

$$N = 5$$

5 4 11 ~~7~~ ~~2~~
+

stergeți cat mai puțin m astfel încât
toate două m. alăturate să nu aibă suma un m. prim



Ștergem MVC

ex 7

Fie $N = (G, s, t, c)$

rețea de transport

f flux în N

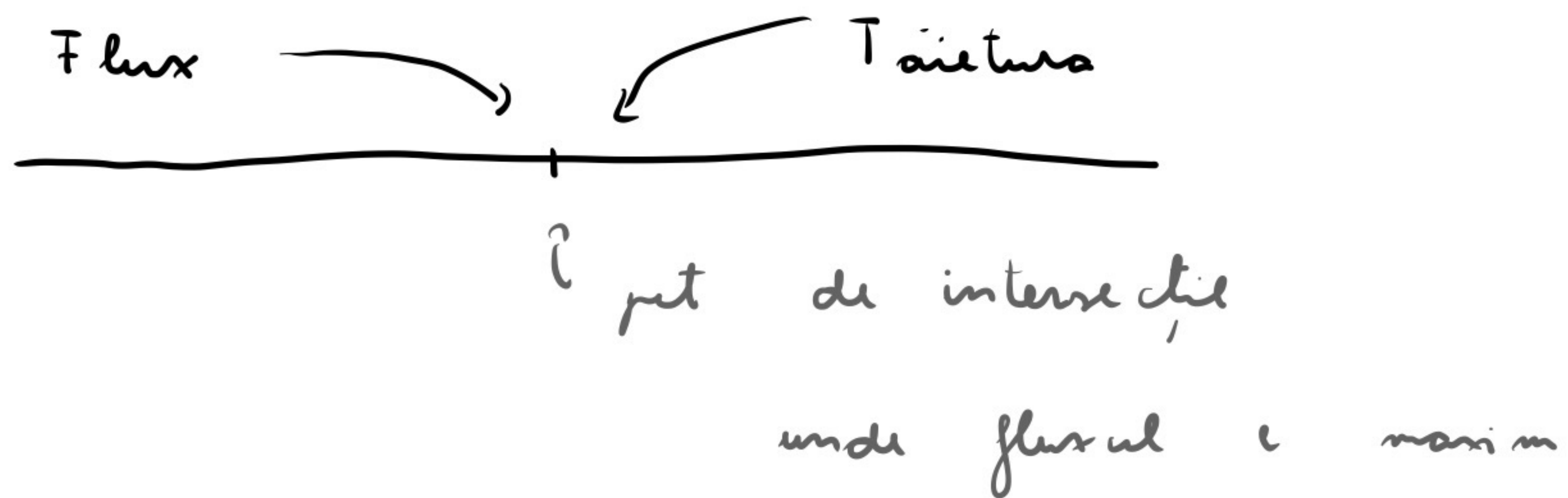
+

True or False

#

a) $f \leq$ suma capacităților care ies din S A

b) $f \geq$ capacitatea minimă a unei tăieturi F



c) $f \leq$ capacitatea maximă a unei tăieturi A

$$F \leq F_{\max}$$

$$C \leq C_{\min}$$

$$F_{\max} = C_{\min}$$

$$\left. \begin{array}{l} F \leq F_{\max} \\ C \leq C_{\min} \end{array} \right\} \Rightarrow F \leq C_{\max}$$